

Rate-Aware Decentralized Detection with Many Sensors

Oren Ram and Ziv Abraham

Department / Institution

City, Country

Email addresses

Abstract—[Abstract placeholder.]

Index Terms—decentralized detection, sensor networks, error exponent, Chernoff information, communication rate

I. INTRODUCTION

[Introduction placeholder.]

II. PROBLEM FORMULATION

Consider binary hypotheses H_1 and H_2 with priors $P(H_1) = p$ and $P(H_2) = 1 - p$. Sensor $i \in \{1, \dots, N\}$ observes $y^i \in \mathcal{Y}$ and applies an encoder $\gamma^i : \mathcal{Y} \rightarrow \mathcal{U}$. Its message $u^i = \gamma^i(y^i)$ is available to a fusion center (FC), which uses $\gamma^0 : \mathcal{U}^N \rightarrow \{1, 2\}$ to produce the final decision $u^0 = \gamma^0(u^{1:N})$.

Under uniform error cost,

$$c(H_j, u^0) = \begin{cases} 1 & \text{if } u^0 \neq j \\ 0 & \text{if } u^0 = j \end{cases} \quad (1)$$

$$J^N(\gamma^{0:N}) \triangleq \mathbb{E}^{\gamma^{0:N}}[c(H^*, u^0)]. \quad (2)$$

Thus, $J^N = P_e$ is the Bayes probability of error. We use the positive normalized error exponent

$$J_{EE}^N(\gamma^{0:N}) \triangleq -\frac{1}{N} \log J^N(\gamma^{0:N}), \quad (3)$$

which is maximized when the error probability decays rapidly.

A. Assumptions

The analysis uses the following assumptions.

- 1) The observations $y^1, \dots, y^N$ are independent and identically distributed conditioned on the true hypothesis.
- 2) Under both hypotheses, the observation laws have densities $f_{y|H}(y|H_j)$ with respect to a common reference measure.
- 3) All sensors have the same finite message alphabet $\mathcal{U}$.
- 4) The likelihood ratio

$$L(y) := \frac{f_{y|H}(y|H_2)}{f_{y|H}(y|H_1)} \quad (4)$$

is regular enough that threshold encoders are well defined; in particular, its derivative is continuous and nonsingular almost surely on the observation space.

- 5) The FC knows the sensor encoders, including any realized common randomization, as well as the received messages.

Under these conditions, the likelihood ratio is a sufficient statistic at each sensor. The results of [1] guarantee an optimal finite-population solution with threshold encoders and a maximum a posteriori (MAP) decoder. They also show that identical independent sensor policies are optimal in the infinite-population problem and asymptotically optimal for large finite populations.

B. MAP fusion rule

For fixed deterministic encoders, define

$$\Delta_N(u^{1:N}; \gamma^{1:N}) \triangleq \frac{1}{N} \sum_{i=1}^N \log \frac{P^{\gamma^i}(u^i|H_1)}{P^{\gamma^i}(u^i|H_2)} \quad (5)$$

The MAP rule is

$$\gamma^{0*}(u^{1:N}) = \begin{cases} H_1 & \text{if } \Delta_N(u^{1:N}; \gamma^{1:N}) \geq t \\ H_2 & \text{if } \Delta_N(u^{1:N}; \gamma^{1:N}) < t \end{cases} \quad t = \frac{1}{N} \log \frac{1-p}{p} \quad (6)$$

III. ERROR-EXPONENT CHARACTERIZATION

For a sensor policy γ^i , define the policy-dependent Chernoff coefficient

$$M(\alpha, \gamma^i) \triangleq \sum_{u \in \mathcal{U}} P^{\gamma^i}(u|H_2)^{1-\alpha} P^{\gamma^i}(u|H_1)^\alpha, \quad 0 \leq \alpha \leq 1. \quad (7)$$

Applying Chernoff bounds to the two MAP error events gives

$$P_e \leq 2p^\alpha(1-p)^{1-\alpha} \prod_{i=1}^N M(\alpha, \gamma^i) \quad (8)$$

Suppose that the sensors use K distinct policies and that a fraction c_k uses policy γ^k , where $c_k \geq 0$ and $\sum_{k=1}^K c_k = 1$. From (8),

$$\liminf_{N \rightarrow \infty} J_{EE}^N(\gamma^{0:N}) \geq -\min_{\alpha \in [0,1]} \sum_{k=1}^K c_k \log M(\alpha, \gamma^k) \quad (9)$$

For an identical policy γ , this becomes the Chernoff information between the two induced message distributions:

$$C_\gamma = -\min_{\alpha \in [0,1]} \log \left(\sum_{u \in \mathcal{U}} P^\gamma(u|H_2)^{1-\alpha} P^\gamma(u|H_1)^\alpha \right) \quad (10)$$

The structural result in [1] therefore reduces the large-network design to selecting one encoder that maximizes C_γ , subject here to a communication-rate constraint.

IV. RATE-AWARE GAUSSIAN POLICIES

Assume

$$y^i|H_1 \sim \mathcal{N}(0, \sigma^2), \quad y^i|H_2 \sim \mathcal{N}(\mu, \sigma^2), \quad (11)$$

with $\mu > 0$. Then

$$\log L(y) = \frac{\mu y}{\sigma^2} - \frac{\mu^2}{2\sigma^2} \quad (12)$$

We compare an always-transmit binary policy with a confidence-based policy that can remain silent, allocating one bit per sensor.

A. Always-transmit policy

For the symmetric threshold $\lambda_V = \mu/2$, let u^1 be sent below the threshold and u^2 otherwise. With $q = Q(\mu/(2\sigma))$, the induced probabilities are

$$\begin{aligned} P(u^1|H_1) &= P(u^2|H_2) = 1 - q, \\ P(u^2|H_1) &= P(u^1|H_2) = q \end{aligned} \quad (13)$$

The Chernoff coefficient is symmetric about $\alpha = 1/2$, and hence

$$C_V = -\log \left(2\sqrt{q(1-q)} \right) \quad (14)$$

Every sensor transmits one bit, so the total rate is $R_V = N$ bits per observation interval.

B. Silent policy

Introduce a no-transmission symbol u^S and symmetric thresholds

$$\lambda_1 = \frac{\mu}{2} - \delta, \quad \lambda_2 = \frac{\mu}{2} + \delta, \quad \delta > 0. \quad (15)$$

The sensor sends u^1 for $y < \lambda_1$, remains silent for $\lambda_1 \leq y \leq \lambda_2$, and sends u^2 for $y > \lambda_2$. Define

$$q_+ = Q\left(\frac{\mu/2 + \delta}{\sigma}\right), \quad q_- = Q\left(\frac{\mu/2 - \delta}{\sigma}\right) \quad (16)$$

The message probabilities are

$$\begin{aligned} (P(u^1|H_1), P(u^S|H_1), P(u^2|H_1)) &= (1 - q_-, q_- - q_+, q_+), \\ (P(u^1|H_2), P(u^S|H_2), P(u^2|H_2)) &= (q_+, q_- - q_+, 1 - q_-) \end{aligned} \quad (17)$$

Again, symmetry gives the minimizing Chernoff parameter $\alpha = 1/2$, so

$$C_S(\delta) = -\log \left(q_- - q_+ + 2\sqrt{(1 - q_-)q_+} \right) \quad (18)$$

The probability of transmission is $\rho(\delta) = 1 - q_- + q_+$ under either hypothesis. If an active sensor sends one bit and silence is detected at no bit cost, then

$$R_S(\delta) = N\rho(\delta) \quad (19)$$

is the expected total rate. Equations (18) and (19) expose the detection–communication tradeoff; δ can be selected by

$$\max_{\delta > 0} C_S(\delta) \quad \text{subject to} \quad R_S(\delta) \leq R_{\max} \quad (20)$$

V. OPTIMAL SILENCE THRESHOLD

At $\alpha = 1/2$, maximizing $C_S(\delta)$ in (18) is equivalent to minimizing

$$M_S(\delta) = (q_- - q_+) + 2\sqrt{(1 - q_-)q_+} \quad (21)$$

Writing $\phi_{\pm} \triangleq \phi((\mu/2 \pm \delta)/\sigma)$ for the standard normal density, $A \triangleq \sqrt{q_+}$, and $B \triangleq \sqrt{1 - q_-}$, differentiating (21) with respect to δ gives

$$\sigma \frac{dM_S}{d\delta} = \frac{(B - A)(\phi_- A - \phi_+ B)}{AB} \quad (22)$$

Since $B > A$ for all $\delta > 0$ (because $B^2 = Q((\delta - \mu/2)/\sigma) > Q((\delta + \mu/2)/\sigma) = A^2$), the sign of $dM_S/d\delta$ is carried entirely by $\phi_- A - \phi_+ B$, so any stationary point of M_S satisfies

$$\frac{\phi_-}{\phi_+} = \frac{B}{A} \iff e^{\mu\delta/\sigma^2} = \sqrt{\frac{1 - q_-}{q_+}} \quad (23)$$

Since $\delta = (\sigma^2/\mu) \log \tau_2$ maps the silence half-width back to the upper likelihood-ratio threshold, the left-hand side of (23) is exactly τ_2 , giving the optimal silence threshold in closed form:

$$\tau_2^{*2} = \frac{P\gamma^S(u^2|H_2)}{P\gamma^S(u^2|H_1)} = \frac{P\gamma^S(u^1|H_1)}{P\gamma^S(u^1|H_2)} \quad (24)$$

Equation (24) reduces the single-parameter search over δ in (20) to evaluating a closed-form condition instead of a numerical line search.

VI. NUMERICAL EVALUATION

For the baseline experiment, use balanced priors and 0 dB SNR: $p = 1/2$ and $\mu = \sigma^2 = 1$. Then $q = Q(0.5) \approx 0.3085$ and

$$C_V \approx 0.0793. \quad (25)$$

Using the optimal thresholds from (24), $\delta^* \approx 0.6170$, i.e. $\tau_1^* \approx 0.5395$, $\tau_2^* \approx 1.8534$, giving $q_+ \approx 0.1320$ and $q_- \approx 0.5466$. Hence

$$C_S \approx 0.1011, \quad \rho \approx 0.5854. \quad (26)$$

This example predicts a larger asymptotic exponent while about 59% of the sensors transmit. A complete evaluation should compare finite- N MAP error probabilities, the limits in (14) and (18), and the rate–exponent curve obtained by varying δ .

VII. SIMULATION

Figure 1 plots the simulated normalized error exponent J_{EE}^N against the number sensors, for both the always-transmit and silent policies, alongside their respective Chernoff-information asymptotes C_V and C_S computed in Section VI. Both curves decay from a large- N -mismatch regime and converge to their asymptote as the sensor count grows, confirming (9) as the correct $N \rightarrow \infty$ limit; the silent policy’s curve lies uniformly above the always-transmit one, consistent with $C_S > C_V$.

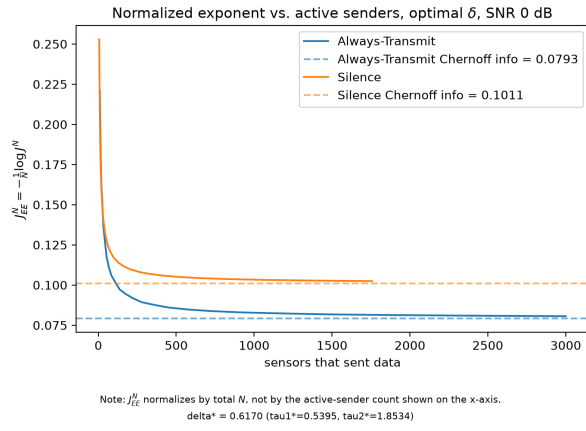


Fig. 1. Normalized error exponent J_{EE}^N vs. number of active senders, SNR = 0 dB, for the always-transmit and silent policies at the optimal threshold $\tau_2^* \approx 1.8534$, with their Chernoff-information asymptotes $C_V \approx 0.0793$ and $C_S \approx 0.1011$ shown as dashed lines.

VIII. CONCLUSION

The proposed structure combines the optimal-policy results for large sensor populations with a rate-aware encoder. Notably, the silent policy attains a strictly larger error exponent than the always-transmit policy while sending less data, showing that selective silence can simultaneously reduce communication and improve detection performance.

REFERENCES

- [1] S. Sanjari, N. Saldi, S. Gezici, and S. Yüksel, “Decentralized detection with many sensors: Optimality of exchangeable and identical encoding policies,” arXiv:2509.21724, Sept. 2025.
- [2] J. N. Tsitsiklis, “Decentralized detection,” MIT Laboratory for Information and Decision Systems, Tech. Rep. LIDS-P-1913, 1989.
- [3] H. Chernoff, “A measure of asymptotic efficiency for tests of a hypothesis based on the sum of observations,” *Ann. Math. Stat.*, vol. 23, no. 4, pp. 493–507, 1952.